

PAPER_1392_RENNINGER_NEGATIVE_RESULT

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PAPER_1392 — UQFF Resolution of the Renninger Negative-Result Experiment (CLOSED — Dispatch Whitepaper)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: B — Quantum Measurement (CLOSED)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "renninger_negative_result"})`

Calculator surface: `calculate_paradox({"paradox": "renninger_negative_result"})`

Closure helper: `_l96_uqff_axiom_renninger_negative_result_closure()`

The Paradox

A wavefunction split into branches by a partial-coverage detector. If the detector does NOT click in time T, the wavefunction collapses into the un-detected branches — a 'collapse without measurement event' apparent paradox.

UQFF Closed Identity

``

`F_U = 1` redistribution fraction = 1.0 EXACT among remaining branches

``

`F_U = 1` global ledger normalization: at the no-click time T, the `F_U = 1` amplitude redistributes 100% into the remaining branches. The 'collapse' is the `F_U` renormalization event, not an interaction.

Physical Interpretation

Renninger's paradox is resolved by recognizing that the `F_U = 1` ledger normalization is the mechanism of 'collapse' — no particle-detector interaction is needed because `F_U` redistribution is a ledger event, not a physical scattering.

Live Calculator Output

```
``python
import uqff_pure_calculator as u
```

```
r = u.calculate_paradox({"paradox": "renninger_negative_result"})["value"]
``
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_renninger_negative_result_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_renninger_negative_result_closure`
- Dispatch keys: `renninger_negative_result`, `renninger`
- Related foundational papers: PAPER_646 ($F_U B_i / F_U = 1$ ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation).

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